CommuteCA: An R package with datasets and standardized methods for job accessibility analysis using Canadian Census microdata

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Abstract

The Canadian Census provides detailed information on the labour force population and how they commute to work. However, the original Census microdata files are confidential and can be accessed only at specially designated Statistics Canada facilities, known as Research Data Centres (RDC). This protocol poses challenges for research reproducibility, since disclosure of at least some of the data in any given analysis will be limited. The objective of this paper is to introduce {CommuteCA}, an R package with analysis ready-data retrieved at a permissible level of aggregation from a RDC, as well as standardized methods for increasing the reproducibility of research conducted at RDCs in Canada. The package combines the characteristics of an open data product and a template for standardized workflows when working at an RDC. The data provided contains information on the labour force by transportation mode and education level, as well as the number of jobs by occupation category, at the census tract level. In addition, CommuteCA provides calibrated impedance functions for all municipalities for 1) four transportation modes (walking, cycling, public transit, and car-motorized vehicles); and 2) education levels of the population. In terms of standardized workflows, the package includes a collection of R Markdown templates for conducting accessibility analyses using the Census microdata files (at an RDC) or the processed datasets (elsewhere), as well as customized functions that facilitate users' analyses.

Keywords: opportunity, employment, impedance, travel, transport

1 Introduction

The Census program in Canada is widely recognized as one of the premier national statistical programs in the world. Data collected are of incalculable value for understanding the status of the country in demographic and socio-economic terms. To ensure that the wealth of information is as widely accessible as possible, Canada's Social Sciences and Humanities Research Council of Canada, Canada's Foundation for Innovation, and Statistics Canada established in the early 2000s the program of Research Data Centres (RDC). The objective of this program is to allow researchers to work with sensitive data as long as work was conducted at a designated Research Data Centre. Before disclosure, Statistics Canada analysts check data requests to ensure that privacy and anonymity of individuals in the population is preserved (Goldmann, 2003). This protocol is eminently sensible but it also poses challenges for open science, since typically any data used in analyses conducted at an RDC will not be available to the public at large.

The objective of this paper is to introduce {CommuteCA}, an `R` package that combines characteristics of an Open Data Product (ODP) (i.e., analysis-ready data) and also includes a set of standardized methods for working with Canada Census microdata files at a Statistics Canada RDC. ODPs result from open and transparent processes that transform raw data to make them ready for analysis (Arribas-Bel et al, 2021). Unlike open data in general, ODPs are characterized mainly by the added value they provide to raw data, but also by their enhanced usability and ease of access. In this respect, {CommuteCA} provides analysis-ready data obtained from Statistics Canada microdata files corresponding to the 2021 Census of Population of Canada (Canada, 2023a), along with a collection of R Markdown templates for conducting accessibility analyses using the Census microdata files (at an RDC) or the processed datasets (elsewhere).

Census data have proved valuable in investigations of urban accessibility, defined as the ease of reaching spatially distributed opportunities (Hansen, 1959), whether by providing land use tables containing the locations of populations and opportunities (Merlin and Hu, 2017; Jasso Chávez and Manaugh, 2025; Cui et al, 2020; Delmelle et al, 2021), or by linking accessibility indicators with individual outcomes, such as their impact on employability levels (Bastiaanssen et al, 2025, 2021), job informality (Boisjoly et al, 2017; Duarte et al, 2023), and job commuting distance (Hu, 2017).

To create {CommuteCA}, we extracted and processed all the data needed to create land use tables. For each Canadian census tract, we counted the number of people in the labour force; the number of people who commute to work by walking, cycling, public transportation, and by car, van, motorcycle, or other automotive vehicles; and the number of employment opportunities. We also provide additional land use tables with the workforce and job opportunities classified by education level, for accessibility analyses that consider potential matches between population scholarly attainment and job position requirements. In addition, we retrieved respondents' commute duration and used them to calibrate impedance functions, categorized by transportation mode and/or education level, for each Census Division (municipality), Census Metropolitan Area or Census Agglomeration, and Canadian province in order to represent the

commuting behaviour of the Canadian population. The R Markdown templates provide a template for conducting all data extraction, pre-processing, and preparation for accessibility analyses for any study area within Canada. These notebooks then are valuable for documenting the workflows implemented, and because they provide standardized methodologies ranging from the very basics of programming in `R` to calculating accessibility models and accessibility maps for transportation planning and policy.

More than ever, accessibility analyses are sparking the interest of planning practitioners in Canada (e.g., Newstead et al (2024)). Therefore, reducing barriers to accessing and using rich but difficult-to-access and complex datasets, such as the Canadian census, is a valuable effort that can contribute to more data-driven decision-making in urban planning.

{CommuteCA} is part of a broader research effort to implement and inspire the best principles of open spatial science (Páez, 2021; Brunsdon and Comber, 2021). As part of this effort, we aim to create ODPs from datasets that, in our perspective, could be more widely used by transportation planners and scholars because they contain valuable variables and for provide high-quality data, often at the national level. {CommuteCA} is released as an `R` package containing code, data, and documentation in a standardized format that can be installed and used by `R` users (including those at RDCs), providing an effective medium for distributing analysis-ready data. This approach has also been demonstrated by other ODPs (Dos Santos et al, 2025; Soukhov and Páez, 2023) and analyses based on them (dos Santos et al, 2026; Soukhov et al, 2023).

In the following sections, this paper discusses the 2021 Census of Population, our main data source, and the process implemented to retrieve and organize into an ODP. Next, we a case study using the datasets and methodological framework to assess the accessibility to employments in the City of Toronto, in order to spark the interest of potential users. This ODP provides not only data and easy-to-use `R` Markdown templates, but also all the code and documentation that make this research project completely reproducible.

2 Canadian Census

The Census in Canada is a quinquennial program that collects important demographic and economic information on the Canadian population to support planning in diverse areas, including public transportation and skills training to improve employment outcomes (Canada, 2023a). The microdata file for the 2021 Canadian Census contains almost 9 million observations and includes almost 900 variables. The data are hierarchically organized into five units of statistical analysis: persons, census families, economic families, households, and dwellings. Each record in the microdata file represents an individual response, which is later grouped into the higher hierarchy levels.

The 2021 Census was Canada's 23rd national census and showed that the Canadian population increased by 5.2% relative to the previous census, reaching almost 37 million people. The majority (75%) of households received a short-form questionnaire covering topics related to language, sex at birth, gender, marital status, and age, while

the remaining 25% received a long-form questionnaire that also covered topics related to activities of daily living, socio-cultural information, mobility, education, and labour market activity. The census data are accompanied by reference guides designed to help users better understand the files. Among these reference guides, the Commuting Reference Guide (Canada, 2023b), along with the Labour Force Reference Guide, contains the variables most relevant for transportation analyses.

2.1 Commute, Labour Force, and Education Reference Guides

The long-form 2021 Canadian Census contains questions related to the Canadian labour force and commuting to work (Canada, 2023b). The labour force consists of people aged 15 years and over living in private households who are either employed or unemployed but actively looking for work. In the labour section, respondents answered questions related to employment status, the number of hours and weeks worked in the previous year, the main reasons for not working the full year (when applicable), and more. Employed respondents were also asked questions related to their industry sector, occupation, employment status (part-time or full-time), and more. Unemployed respondents were asked about the main reason for being unable to find work, the reason for not accepting a new job (when applicable), and whether they planned to start a new job (when applicable).

In the section related to commuting, the questions referred to the journey to work. These questions include whether respondents worked at home (including farms), outside Canada, or had a fixed workplace address. Other questions include the workplace location, main mode of commuting, other commuting modes, vehicle occupancy, commuting duration, and departure time.

According to the 2021 Canadian Census, about 30.3 million Canadians were aged 15 years or older and living in private households, making them eligible to participate in the labour force (about 82% of the population). However, the labour force participation rate was 63.7% (about 19.3 million people), meaning that approximately 11 million people were unemployed but were not actively seeking work due to early retirement, education, family care responsibilities, illness or disability, personal choice, or because they were discouraged workers - cases in which a person wanted a job but stopped looking because they believed no work was available or could not find suitable employment in their local labour market. Among those in the labour force, the unemployment rate was about 10.3%. The 2021 Census revealed that the labour market was still being affected by the COVID-19 pandemic, with the number of people in the labour force showing a slight decline, from 19.5 million in 2016 to 19.3 million.

The number of employed people was approximately 17.3 million. Of that total, 4.2 million worked from home, representing 24.3% of the employed workforce - a share three times higher than the proportion observed in 2016. The 2021 Census also revealed that approximately 13 million people commuted to work. Among them, the majority (84%) commuted by car, truck, or van; 8% used public transportation; 5% walked; and 1% cycled. In terms of commuting duration, 32% of commuters travelled for less than 15 minutes, 35% between 15 to 29 minutes, 19% between 30 to 44 minutes, 7% for 45 to 59 minutes, and 7% for 60 minutes or more. For Canadians aged 15 years and over, 16% do not have a certificate, diploma, or degree, while 27% have a high

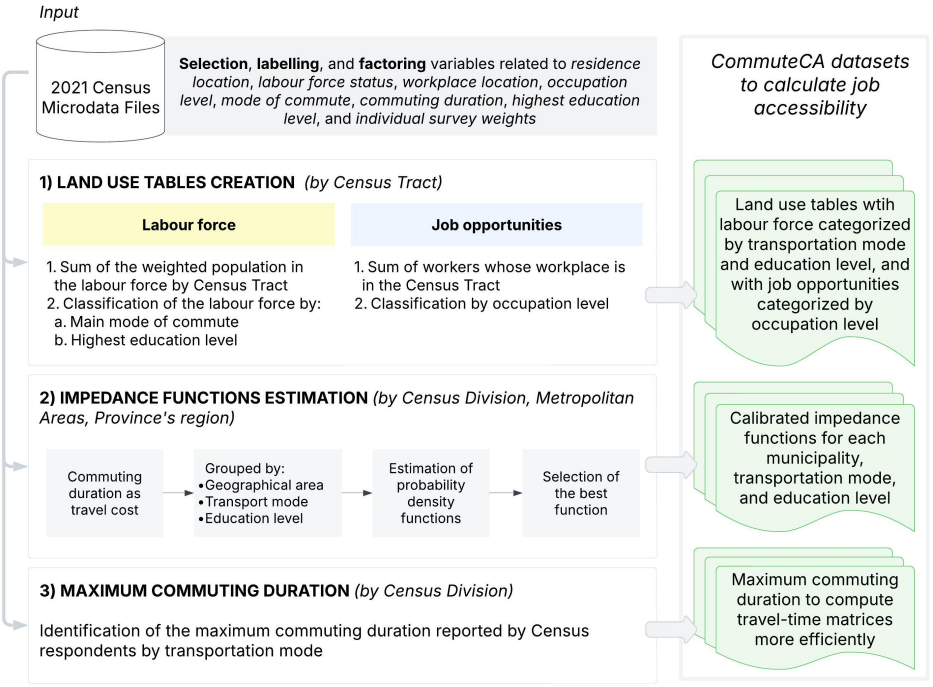


Fig. 1 CommuteCA datasets obtained from the 2021 Canadian Census microdata files.

school diploma or equivalency certificate as their highest level of education, and 57% have earned at least a postsecondary certificate.

3 Data process

Figure 1 presents a diagram illustrating the processes applied to the Census microdata files to create the {CommuteCA} datasets. These datasets were created with the intention of providing all the data required to calculate job accessibility for any study area in Canada.

Acessibility refers to the ease of reaching spatially distributed opportunities (Hansen, 1959) and results from the interaction between land use and transportation systems (Geurs and van Wee, 2004). Based on this, we grouped the inputs required for these calculations into three categories: 1) a land use table reporting, for each spatial unit, the number of opportunities - in this case, employment opportunities -, and the number of people competing for these opportunities when competition is considered by the accessibility measure; 2) a routing table containing the travel costs between each origin and destination, typically expressed as travel-time matrices for one or more transportation modes; and, 3) one or more impedance functions representing the difficulty of accessing opportunities. {CommuteCA} provides the first and third of these

components, while also proving inputs to generate the second using routing engines such as {r5r} (Pereira et al, 2021).

We begin by creating land use tables for all areas containing a Census Tract (CT). CTs are intra-municipal spatial units with populations ranging from 2,500 to 7,500 people, with an average of 5,000, and are located in Census Metropolitan Areas (CMAs) and Census Agglomerations (CAs) that had a population greater than 50,000 at the previous census (2016). Although CTs do not cover the entire country, 84% of the Canadian population lived in a CMA or CA (over 31 million people). CTs are distributed across all Canadian provinces, except Prince Edward Island, whose total population was only slightly more than 154,000 in 2016.

In terms of the labour force, {CommuteCA} contains land use tables with data on the total labour force in each CT, as well as the labour force classified by highest education level and/or transportation mode. First, we summed the weighted population within the labour force in each CT. Then, we identified the main transportation mode used to commute to work, when applicable, and classified it into four categories: "Car/Motorized" (car, truck, van, or motorcycle), "Public Transit" (bus, subway, rail, streetcar, train, or passenger ferry), "Walk", and "Bike". For workers who worked remotely or did not have a fixed workplace, as well as unemployed people, we used the category "Other or non-commuter". Next, we classified members of the labour force based on their highest education level into three categories: "High school or no certificate", "College or apprenticeship certificate" to represent postsecondary certificate or diploma below university level, and "University certificate or higher".

For job opportunities, we summed the number of workers whose reported workplace was located in each CT and used this total as an estimate of the number of employment opportunities within the CT. We also classified these job opportunities according to the 2021 National Occupational Classification (NOC) and its Training, Education, Experience and Responsibilities (TEER) categories. As a result, the tables provide estimates of the number of "Management and professional" jobs, corresponding to management and professional occupations that generally require completion of a university degree or previous experience and expertise in subject-matter knowledge from a related occupation (TEER 0 and 1); "College or apprenticeship", referring to occupations in TEER 2 and TEER 3, which usually require completion of a postsecondary education program or apprenticeship training program; and "High school or short-term experience", referring to occupations that usually require completion of secondary school or a short work demonstration, with no formal educational requirements.

When generating travel-time matrices using routing engines such as {r5r}, analysts must specify a maximum trip duration - that is, the longest travel time (in minutes) considered during route calculation - to avoid unnecessary computations. For each Census Division (municipality), regardless of whether it is located within a CMA or CA, we identified the maximum commuting duration reported by Census respondents.

Next, we estimated the best impedance functions to represent the commuting behaviour of Canadians. We adopted Probability Density Functions (PDFs) in which an impedance function can be interpreted as the probability density of a trip occurring for each value of travel cost duration, following the approach of Soukhov and Paez (2024), Šveda and Madajová (2023), and dos Santos et al (2026). We applied

the {fitdistrplus} package (Delignette-Muller and Dutang, 2015) to calculate the best-fitting PDF for every Census Division (municipality), CMA, and Province, transportation mode (walking, cycling, public transit, and car/motorized) and/or education level, by modelling commuting duration as travel cost and testing uniform, negative exponential, gamma, normal, and lognormal distributions. Then, we selected the best impedance function based on the lowest Akaike Information Criterion value (Akaike, 1974).

In addition to these datasets that enable calculating job accessibility for CMAs or CAs in Canada, we also created testing datasets based on the 2021 Census of Population. These testing datasets were synthetically generated and contain variables for accessibility analyses to make it possible for analysts to practice coding outside Statistics Canada offices or RDCs, since the access to the census microdata files is restricted. One of these datasets has 250,000 observations distributed across Canada and contains 17 variables, such as respondent identifiers and sample weight, geographical information, status in the labour force, and commuting to work variables. The second dataset represents respondents living in the City of Toronto, contains over 50,000 observations and 32 variables, that in addition to the previous set of variables, also includes sociodemographic variables that can be used to test regression models, such as income, race, and gender.

All code used to generate the {CommuteCA} datasets is publicly available through the project's GitHub repository, allowing researchers to reproduce the datasets, verify the implementation, identify and correct potential errors, and adapt the workflows to their own applications.

3.1 Research data centre vetting process

All datasets included in {CommuteCA} were subjected to the confidentiality rules for the 2021 Census of Population through a vetting process conducted at the RDC. To be released, {CommuteCA} datasets had to comply with rules related mainly to minimum geographic area sizes, statistical outputs, and counts. According to the vetting rules, statistics must not be released for identifiable geographic areas containing fewer than a minimum number of persons. For place-of-work data, which are the source of the data used to produce all our datasets, the population under consideration is the employed labour force.

In addition, all statistics other than counts must be based on weighted data and must be calculated from at least a minimum number of respondents whose combined weights must also exceed a minimum threshold to be releasable. These statistics must also follow a specified rounding system that varies according to sample size. For counts, all values must be weighted and rounded to the nearest multiple of five.

4 Standardized methodologies and fuctions

Having provided the datasets required to calculate accessibility models, {CommuteCA} also provides a collection of `R` Markdown templates containing the code (including personalized functions) needed to estimate these models and assess the relationship between accessibility and employment outcomes. Therefore,

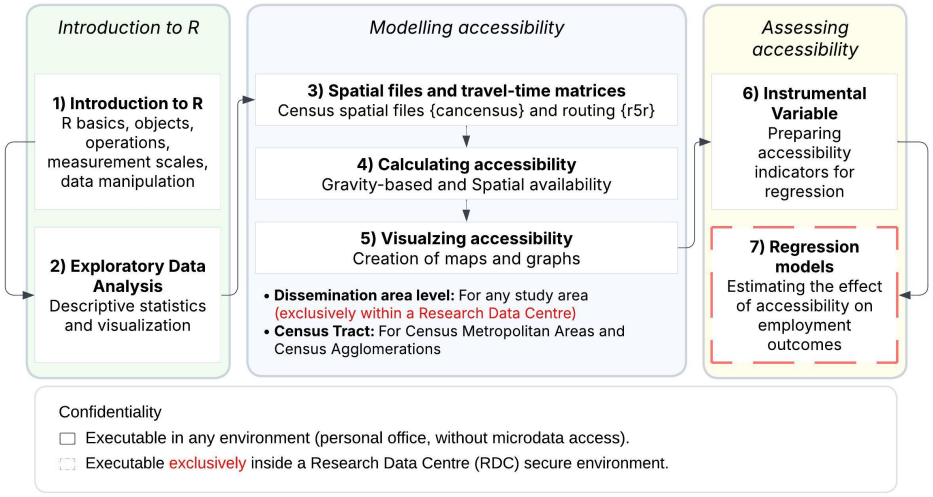


Fig. 2 R Markdown templates available in CommuteCA to perfom accessibility analyses.

the package provides a transparent and standardized methodology for conducting transportation analyses using the 2021 Canadian Census.

Figure 2 presents a diagram illustrating the R Markdown templates available in {CommuteCA}. As shown in the figure, these templates enable analysts to estimate accessibility models at the CT level for multiple CMAs and CAs without requiring special access to the census microdata files. In addition, they allow analysts to model job accessibility for any study area in Canada, including areas outside metropolitan areas, at the Dissemination Area (DA) level - a spatial unit smaller than the CT, with a population between 400 and 700 people, which was not included in the package datasets because of the confidentiality restrictions defined by Statistics Canada.

In total, {CommuteCA} contains a couple of `R` Markdown templates that guide analysts through an accessibility analysis, from an introduction to `R` to the identification of accessibility thresholds. They are divided into three groups: Introduction to `R` (Templates 1 and 2), Accessibility Modelling (Templates 3 to 5), and Assessing Accessibility (Templates 6 to 7). The first template provides an introduction to the `R` language. It also introduces literate programming, data objects, basic operations, data measurement scales, and data manipulation techniques in `R`. The second template introduces Exploratory Data Analysis (EDA), demonstrating how to produce descriptive statistics and create appropriate visualizations for different data types.

The accessibility modelling templates begin with the third template, which guides analysts in organizing census spatial data and demonstrates how to obtain census spatial files using the {cancensus} package and how to generate travel-time matrices using the {r5r} routing engine. The forth template explains how to calculate job accessibility using the traditional Hansen accessibility measure (gravity model) and the spatial availability model proposed by Soukhov et al (2023). The fifth template explains how to plot maps to visualize job accessibility.

The remaining templates focus on assessing accessibility outcomes using the accessibility models generated in the previous stage. The sixth template demonstrates how to create spatial filters for accessibility indicators so that they can be incorporated into regression models estimating the effects of accessibility on the probability of employment and labour force participation. Finally, the seventh template demonstrates how to estimate these regression models, including the processing of socioeconomic, demographic, and environmental variables.

The package also includes customized functions that allow users to select the best impedance function for a study area, plot theoretical impedance values, calculate accessibility indicators, and create spatial filters for the accessibility indicators.

5 Results

5.1 {CommuteCA} data sets

{CommuteCA} contains labour force and job opportunity estimates for 6,002 CTs, representing 96% of the 6,247 CTs in Canada. Almost 14.7 million people in the labour force are represented in the included CTs, representing around 77% of the Canadian labour force. The labour force not included in the dataset is composed of people who do not live in metropolitan areas or who live in CTs excluded during the Statistics Canada vetting process. Around 53% of the labour force commuted by car-motorized modes, followed by 6.5% who commuted by public transportation. Active commuters (walking or cycling) represented less than 4% of the labour force, while commuters who used non-specified modes and non-commuters, due to being unemployed or because they were working from home, represented 37% of the labour force. Due to this high share of the labour force not commuting to work, aggravated by the COVID-19 pandemic, we also made available datasets with the non-commuter labour force distributed across the other transportation modes, according to the share of these modes in each CT. In terms of education, the majority of the labour force had a certificate or degree above high school, with 40% having a university certificate or higher and around 28% having a college or apprenticeship certificate.

Almost 70% of the job opportunities in Canada are represented in the {CommuteCA} datasets (around 11.8 million). Again, the jobs not included in the dataset are either located in CTs excluded during the vetting process or outside metropolitan areas. According to the TEER categories, around 38% of these job opportunities are categorized as management or professional positions, which are usually occupied by workers with at least a university degree or certificate; around 35% refer to positions that require college or apprentice certificates, and the remaining 27% are positions for which no certification or only a high-school certification is required.

In most cases (80%), the municipalities have impedance functions for all four transportation modes. In some cases, the package does not include an impedance function because there were no minimal labour force respondents who reported commuting using a given mode in the sample. For almost 15% of the municipalities, only three impedance functions were provided, while for less than 6% only two functions were provided. We were able to estimate impedance functions for car/motorized travel

and walking in all municipalities, while functions for public transit and cycling were estimated for 90% and 84% of municipalities, respectively.

{CommuteCA} contains maximum travel times and impedance functions for all 293 Census Divisions in Canada. The maximum travel time identified was 240 minutes (4 hours) for all four transportation modes: car/motorized, public transit, walking, and cycling. The shortest maximum travel time identified was 5 minutes for cycling in Les Etchemins, Quebec. For the other modes, the shortest maximum travel times were 15 minutes for walking (identified in Guysborough, Nova Scotia), 30 minutes for public transit (identified in Communauté maritime des Îles-de-la-Madeleine, Quebec), and 90 minutes for car/motorized travel (identified in Kitikmeot, Nunavut).

5.2 Case study

This section presents an application of the {CommuteCA} `R` package using a selected study area. The application demonstrated in this case study can be conducted by any user who installs the package, without requiring access to the restricted microdata files of the 2021 Canadian Census. Installation instructions of the {CommuteCA} `R` package are available in the project's GitHub repository, and all analyses presented in this section were conducted using the `R` Markdown templates provided with the package.

For this demonstration, we will realize a multimodal accessibility analysis for the City of Toronto, in Ontario, Canada (Figure 3). According to the 2021 Canadian Census (Canada, 2023a), almost 2.8 million residents live in this area, which covers approximately 631 square kilometres.

Having defined a study area, we can use one of the {CommuteCA} R Markdown templates to download spatial files using {cancensus}. Next, using {r5r}, we can generate routes for the selected transportation modes and obtain travel-time matrices representing the estimated travel duration between each origin and destination. These matrices are limited by the maximum travel times for each transportation mode and Census Division available in {CommuteCA}. Both the methodology for downloading spatial files and the methodology for generating travel-time matrices are standardized within the collection of templates.

The {CommuteCA} dataset contains 582 of the 585 CTs available in the study area, with the three missing CTs excluded because they were subject to Statistics Canada vetting rules. According to our dataset, the study area has a estimated labour force of around 1,518,405 people and approximately 1,258,455 job opportunities. According to our estimates, the majority (61%) of workers who commute to work in the study area are car commuters, followed by public transit users (about 27%), pedestrians (almost 10%), and cyclists (less than 3%). Figure 4 presents a map of the estimated labour force population density (per km^2) by transportation mode, created using the {CommuteCA} datasets. It can be seen that the south-central part of the city concentrates a larger active population, especially walkers, while public transit users and, especially, car commuters are spread throughout the city.

Each transportation mode contains its specific impedance function, calibrated using only commuters from the City of Toronto. When calibrated using empirical data, impedance functions illustrate the distribution of observed commuting travel

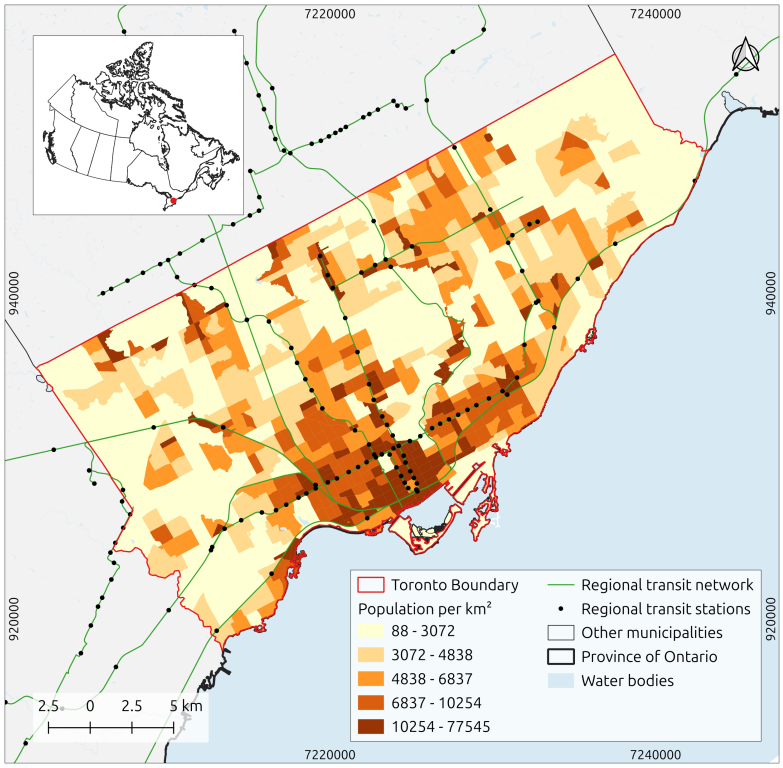


Fig. 3 Population density of the City of Toronto (study area).

behaviour across different travel costs. The analysis of these functions generally reveals a distance-decay effect, in which greater distances between origins and destinations lead to lower movement intensity and a lower probability of observing trips. However, the shape of this distance-decay effect varies across transportation modes and municipalities, showing differences in population behaviour when commuting to work.

Table 1 Calibrated impedance functions for the City of Toronto (Census code 3520).

Mode	Impedance function	Parameter 1[1]	Parameter 2[1]
Walk	Gamma	1.330	0.085
Bike	Log-normal	2.939	0.648
Public transit	Gamma	3.601	0.079
Car-motorized	Gamma	2.329	0.086

[1]For each probability distribution, Parameter 1 and Parameter 2 correspond to their standard defining parameters: the mean and standard deviation (for lognormal) and the rate and shape (for gamma).

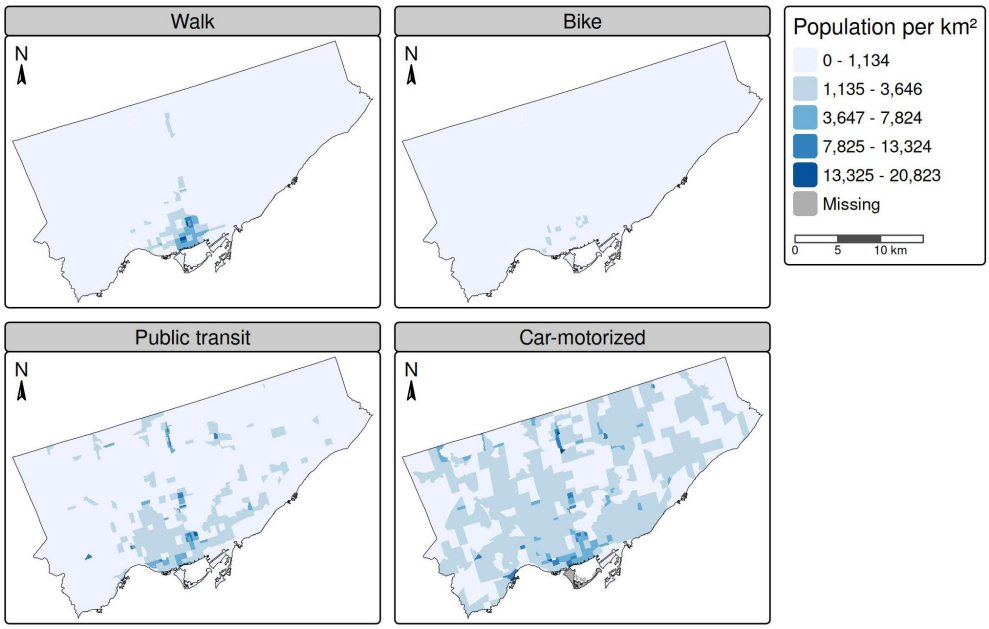


Fig. 4 Estimated labour force population density by transportation mode.

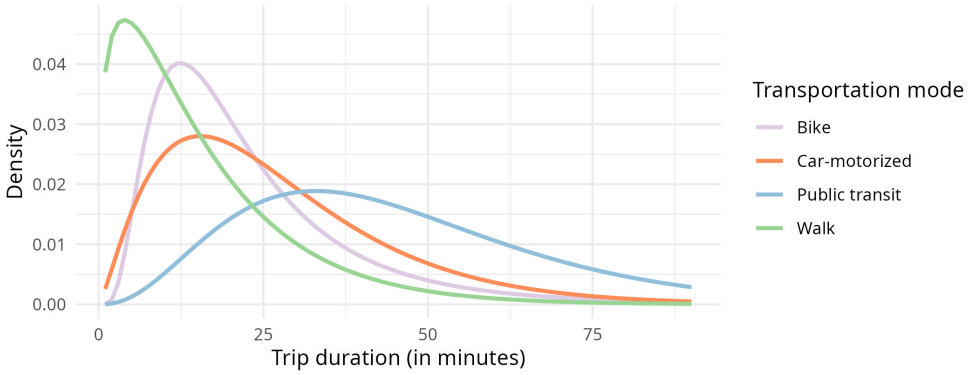


Fig. 5 Toronto calibrated impedance functions by transportation mode.

Table 1 shows the calibrated impedance functions for the study area, while Figure 5 displays their distributions by transportation mode. Almost all trips are better represented by gamma distributions, except for bike trips, which are better represented by a lognormal distribution. In all cases, the distributions show that at durations close to zero minutes, the trip density is lower (with a density of zero for public transit and bike trips). After a few minutes, the density reaches a peak, followed by a decline toward zero at very high travel durations, indicating a low probability of observing trips at these travel times.

For walking trips, the probability density peak occurs at around 4 minutes. For cycling trips, the peak occurs at 12 minutes, and a similar behaviour is observed for car commuters, although the peak occurs at 15 minutes and there is a greater probability of trips continuing to occur beyond 50 minutes. Finally, public transit users show the longest commuting durations, with a lower peak density than the other modes and curves that are more distributed across longer durations. The peak occurs at 33 minutes. From a transportation planning perspective, these differences among impedance functions highlight how using a generic function in accessibility studies can lead to distorted accessibility indicators, as opportunities may be weighted incorrectly according to the observed travel behaviour of different populations and transportation modes.

At this point, the analyst has all the data required to calculate job accessibility models. Using the {CommuteCA} templates, the analyst can calculate job accessibility and create maps such as those displayed in Figure 6.

The maps on the first column show the result of a gravity-based model, as proposed by Hansen (1959), while the maps on the second column show the absolute spatial availability and the spatial availability per person, both proposed by Soukhov et al. (2023; Soukhov et al, 2024). These models differ because the Hansen (1959) model does not consider competition between people for available opportunities. It has the advantage of being simple and well established in transportation analyses. However, its output is not directly interpretable as a number of accessible jobs, making the indicator more difficult to explain. In contrast, spatial availability, the measure proposed by Soukhov et al (2023) has the advantage of being constrained, meaning that the model output can be interpreted as the number of jobs accessible to people in each spatial unit. If we sum the accessibility values across all spatial units, we obtain the total number of jobs available in the system. In addition, because it considers competition, it prevents a single job opportunity from being counted as accessible to more than one person.

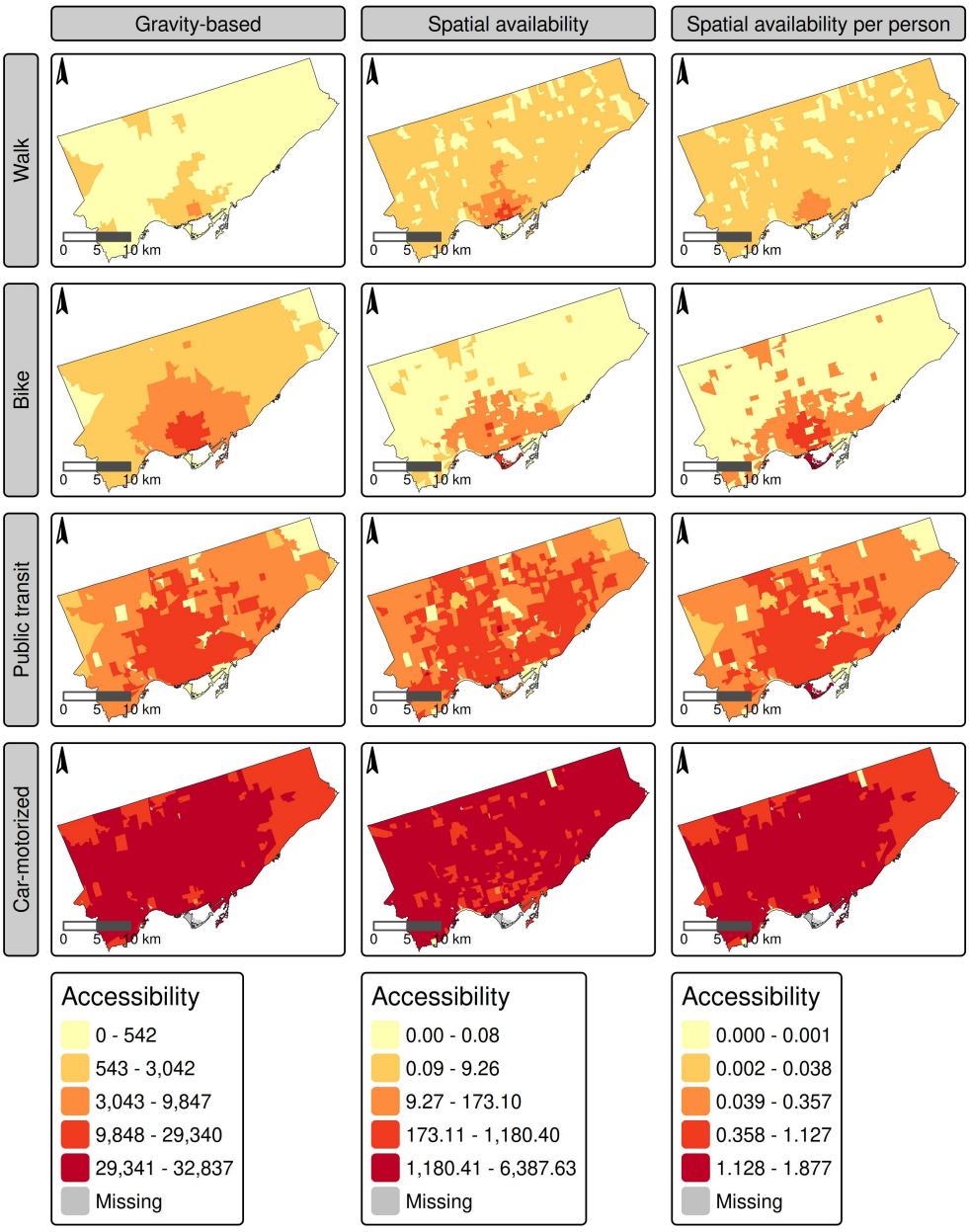


Fig. 6 Job accessibility indicators by transportation mode for the City of Toronto.

Lastly, the results shown in the third column of Figure 6 refer to the spatial availability per person, which refers to the amount of jobs spatially available divided by the number of people competing for those jobs. This last model is suitable for situations where the spatial unit (e.g., CTs) does not have a standardized format, such as

regular grids, allowing the comparisons between areas with different population sizes. As observed for the other measures, the spatial availability per person shows that, generally, car commuters can access substantially more jobs than users of other transportation modes, without large variation across locations within the study area. In fact, 86% of the jobs available in the study area are accessible to people who commute by car, although car commuters represent 61% of the study area's labour force (see Figure 7). The results for walking and cycling, however, show that higher levels of accessibility are generally concentrated around downtown areas. For public transit users, as expected, higher accessibility is observed in areas with more public transit infrastructure, such as subway and regional rail lines (see Figure 3).

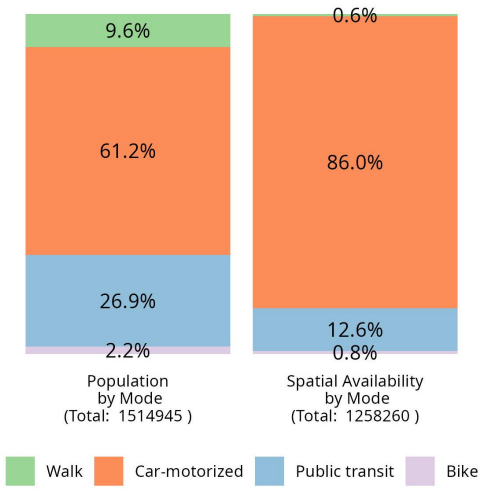


Fig. 7 Share of labour force and jobs spatially availables by transporation mode for the City of Toronto.

5.3 Processing 2021 Canadian Census Microdata Files with {CommuteCA}

For researchers with access to the restricted microdata files of the 2021 Census of Population, it is also possible to conduct the analyses shown in our case study at the DA level. Figure 8 shows the spatial availability measure at the DA level for the City of Toronto. As shown in the figure, the spatial pattern does not differ substantially from that observed at the CT level, but for some applications, a more geographically detailed indicator may be preferable. In addition, the DA level makes it possible to conduct analyses for any study area in Canada, which may be relevant for rural and remote areas where CTs are not available. However, analysis at the DA level is more likely to have cases excluded during the RDC vetting process, as we can see from the number of DAs without accessibility information. This happens because DAs are smaller geographical areas with lower population sizes when compared to CTs, and are more prone to failing to meet the population size requirements. As a consequence,

although the high spatial detail of the DA allows for more refined analysis, these analyses are more likely to be affected by exclusions during the vetting process.

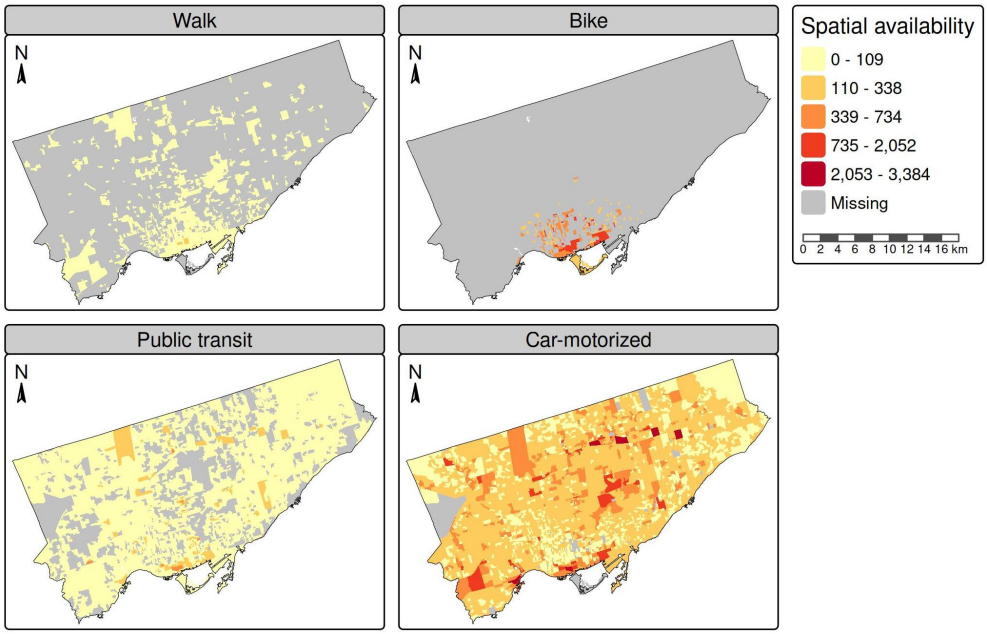


Fig. 8 Job spatial availability in Toronto at the Dissemination Area level.

In addition, {CommuteCA} enables researchers to extend the accessibility analysis by linking accessibility indicators to socioeconomic and demographic variables. The collection of `R` Markdown templates includes instructions for estimating different regression models to examine how job accessibility is associated with two employment outcomes: employment status (i.e., the probability of being employed) and, among those who are unemployed, the probability of having looked for a job during the four weeks preceding the 2021 Census interview.

For all `R` Markdown templates in which census microdata files are processed, the methodologies include a final code section dedicated to the Statistics Canada vetting process for RDCs. This section produces an output formatted according to Statistics Canada rules for releasing analyses and data derived from census microdata, including comments and colour-coded spreadsheets to facilitate the verification process by the RDC analyst while helping protect the confidentiality of census respondents.

6 Concluding remarks

Statistics Canada Microdata Files offer a wealth of information that is usable only at designated facilities called Research Data Centres. Obtaining permission to work at one of thse facilities often entail a lengthy approval process, which makes it difficult

for urban planners and transportation researchers to make data-driven decisions. Even when access is granted, preparing the data for analysis is not straightforward, given its size and complexity. Moreover, the process of extracting information of interest from surveys like the Census is time-consuming, technically challenging, and prone to error, particularly for analysts without extensive experience working with these files.

To reduce the friction in terms of using this invaluable data source, we created {CommuteCA}, an ODP with analysis-ready data and standardized methodologies for job accessibility analyses in Canada using data from the 2021 Census of Population. In the form of an `R` package, {CommuteCA} was developed after collecting, cleaning, and processing the confidential microdata files of the most recently released Canadian Census, providing information on the labour force (classified or not by transportation mode and education level) and the number of jobs (classified or not by occupation category) for all Census Metropolitan Areas and Census Agglomerations in Canada with census tracts. In addition, the package provides standardized methodologies describing how to conduct accessibility analyses using the Census microdata files or the already processed datasets, in addition to customized functions that facilitate users' analyses.

This package comes to join other initiatives, such as package {cancensus} (von Bergmann et al, 2026), which can be used to retrieve analysis-ready data frames and spatial data in multiple formats, as well as convenience functions for working with Canadian census data. By providing data and tools to work with microdata files, {CommuteCA} aims to support urban accessibility analyses. To our knowledge, no existing package has yet released the data and standardized methodology required to support urban accessibility analyses for any study area in Canada. In particular, existing packages do not provide workplace locations at the intra-urban level or impedance functions that represent population travel behaviour when commuting to work, nor do they provide a standardized methodology for conducting accessibility analyses.

The package allows users to conduct job accessibility analyses for any study area in Canada at the Census Dissemination Area level, provided that the analyst has been granted access to the Census microdata files, and conduct accessibility analyses for any study area within a Census Metropolitan Area or Census Agglomeration (where about 84% of the Canadian population resides) at the Census Tract level. Similar to other ODPs, we believe that the value of {CommuteCA} lies in its transparency, accessibility, and ease of use, which facilitates the addition of complementary datasets in the future.

With {CommuteCA}, we aim to contribute to the academic community by promoting transparent research practices that encourage replication and innovation in related fields. We believe that this package will serve as a basis for new research on not only on job accessibility, but will help to inspire similar standardized approaches to work with other datasets among the trove of information that is currently available only at RDC facilities.

Declaration of Competing Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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In addition, we would also like to thank some other `R` packages that were essential to the production of {CommuteCA}, namely: {cancensus} (von Bergmann et al, 2026), {r5r} (Pereira et al, 2021), {fitdistrplus} (Delignette-Muller and Dutang, 2015), {tmap} (Tennekes, 2018), {sf} (Pebesma, 2018), and packages within the {tidyverse} initiative (Wickham et al, 2019) - especially {ggplot2} (Wickham, 2016), {dplyr} (Wickham et al, 2026), and {tidyr} (Wickham et al, 2025).

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Data availability statement

The {CommuteCA} `R` package can be found and installed on Github.

Open science statement

We used literate programming to develop this paper, making the data analysis code available in the same Github. repository. Together, these practices help ensure the transparency and reproducibility of our study, facilitate collaboration and further analysis, and align with best practices in open spatial data science (Páez, 2021; Arribas-Bel et al, 2021).

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